Topic: Cyber-Safe Education: Building a Secure and Responsible Digital Learning Ecosystem for Viksit Bharat

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1. One Wrong Letter

Recently a message started circulating among my friends: a link for one-month free Discord Nitro. I've seen more than five people I know click on it. Each arrived on a page that looked exactly like Discord's login screen, typed in their credentials and within seconds lost control of their account. The clever bit: the same link was then forwarded to each friend on the list, through each hijacked account. The victims had become vehicles for delivery.

When the link came to me, I saw what the problem was in seconds. The domain was not discord.com but a lookalike, something like *discored.com*. One extra letter. I saw it not because I am smarter than my friends. I saw it because after spending considerable time studying cyber security, I had been trained to look at the part of the screen most people never bother to read.

That difference, between what we are taught to look at and what attackers exploit, is the central problem cyber-safe education must solve if our country's digital learning ecosystem is to carry us towards Viksit Bharat.

2. The Tip of the Iceberg

Ask students what cybersecurity means, and the answers are quite predictable: fraud calls, phishing emails, fake promotions. These dangers are real. The NCRB recorded 1,01,928 cybercrimes in 2024, the first time India crossed one lakh; 72.6% were frauds. But they are only the visible tip. Beneath the waterline lie the things that decide whether an attack succeeds: social engineering, the protocols that carry our data, the frameworks attackers follow, and the code we ourselves write.

The waterline matters more every year. More than 26 crore students now hold APAAR academic IDs, and DIKSHA, SWAYAM and the National Digital Library of India, built at IIT Kharagpur

here in West Bengal, carry our learning. Education is also among India's most attacked sectors: Check Point reports Indian educational organisations faced 7,684 attacks per week on average in 2025, more than twice the national average. Awareness of the tip is no longer enough.

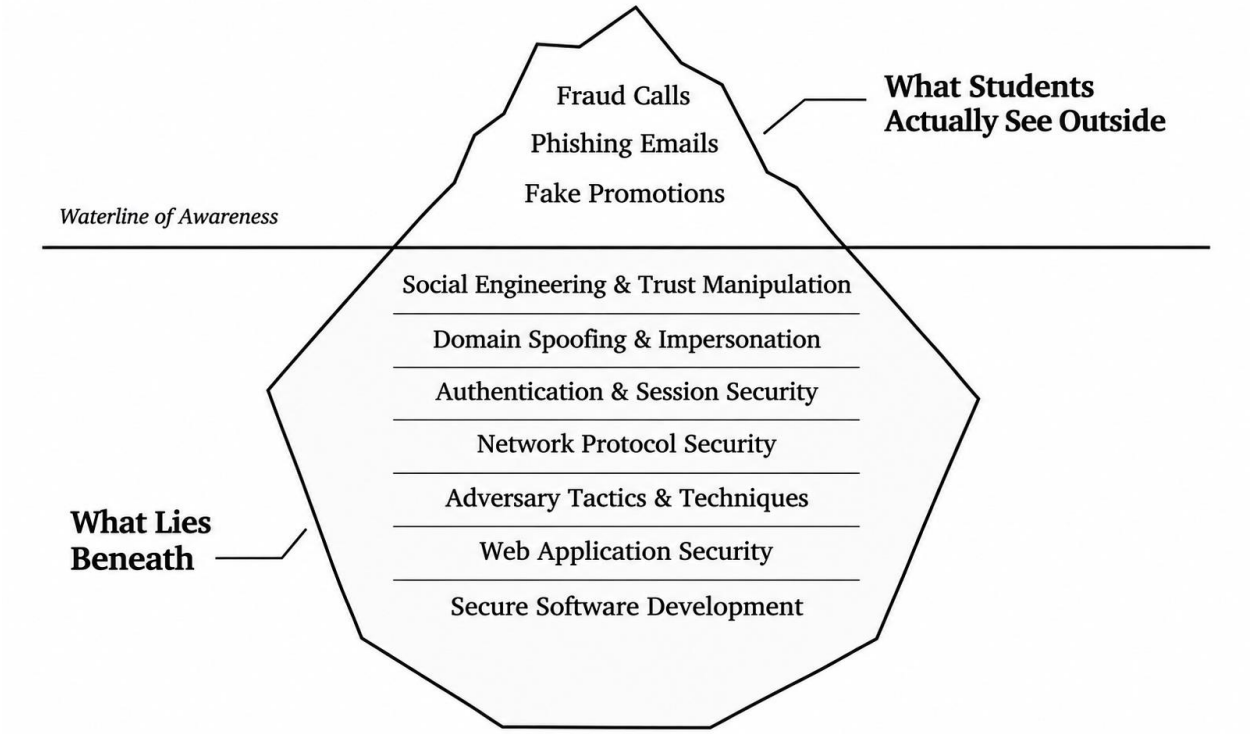


Fig 1. The iceberg of cyber-safety: what students see versus what decides an attack.

3. Anatomy of One Attack

Though it looks like a simple fake offer, the incident maps neatly onto MITRE ATT&CK, the global knowledge base of real adversary behaviour (Table 1).

Resource development	Attacker registers a lookalike domain	T1583.001 Acquire Infrastructure: Domains
Initial access	"Free Premium" link delivered	T1566.002 Phishing: Spearphishing Link
Credential access	Fake login page captures credentials or session token	T1056.003 Web Portal Capture; T1539 Steal Web Session Cookie
Account control	Attacker logs in as the victim	T1078 Valid Accounts
Lateral movement	Same link sent from the victim's trusted account	T1534 Internal Spearphishing

The message came from a friend, so the attacker was not asking for trust, instead, they were borrowing it. A message from your best friend doesn't seem harmful. It also demonstrates why one-time codes alone may not save us: if a fake page captures a code or session token in real time, the attacker simply walks through the door the victim opened.

4. The Logic Behind One Wrong Letter

In order to understand why this happened so fast, we can leverage two mathematical tools: Levenshtein edit distance, which explains why the trap was hard to spot, and the Galton–Watson branching process, which explains why it spread so rapidly.

4.1 The lookalike domains

"discord" → "discored" has a Levenshtein (edit) distance of 1: a single inserted letter. For a seven-character name over the 37 characters allowed in a domain label (a–z, 0–9 and the hyphen), the distinct distance-1 variants number about:
7 deletions + (7 × 36) substitutions + (8 × 37 − 7) insertions = 7 + 252 + 289 = **548**.

That is nearly 550 lookalikes before counting other top-level domains or Unicode lookalike characters. The defender must watch hundreds; the attacker needs only one.

4.2 The spread chain

Let each hijacked account message f friends, of whom a fraction p click. Each victim then produces on average $k = f \cdot p$ new victims, and after n rounds the total is

$$T_n = 1 + k + k^2 + \ldots + k^n = \frac{(k^{n+1} - 1)}{(k - 1)}$$

With $f = 150$ and $p = 2\,\%$, $k = 3$, and four rounds give $1 + 3 + 9 + 27 + 81 = 121$ accounts. If education lowers the click rate to $p = 0.5\,\%$, then $k = 0.75$ and the expected size of the entire chain is $\frac{1}{(1-k)} = 4$. In Galton–Watson branching theory, a chain with $k \leq 1$ dies out with certainty. (Simplified model: all accounts assumed alike.)

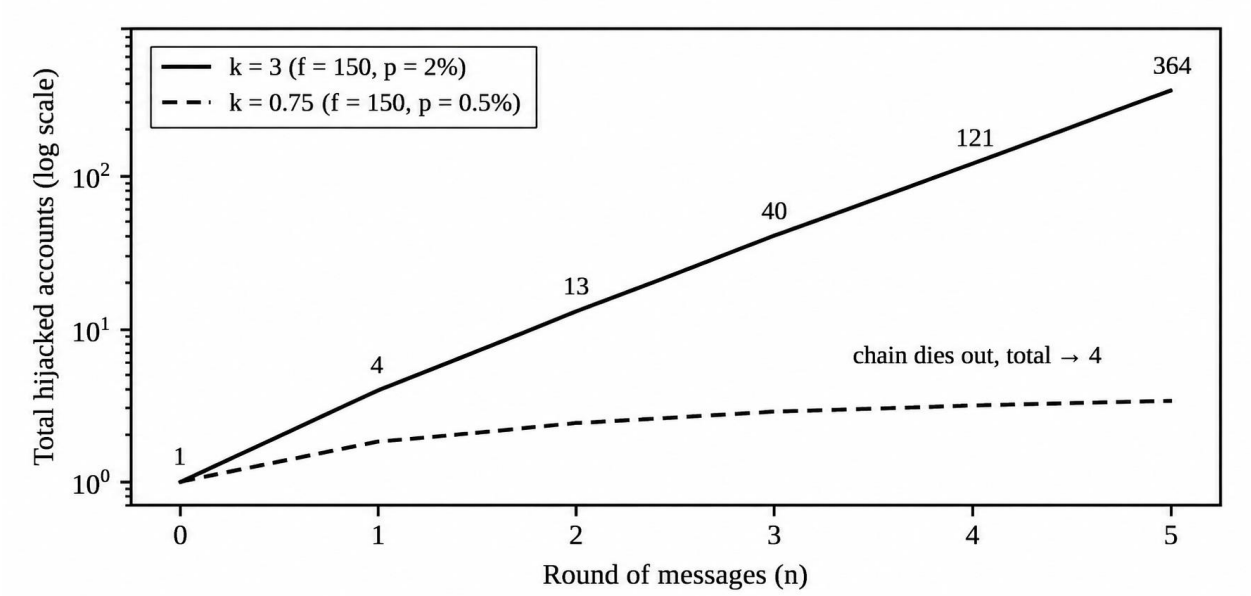


Fig 2. Accounts hijacked per round when k = 3 versus k = 0.75.

5. The Builder's Blind Spot

My second observation is a closer look to my environment. Many of my peers can build a complete website by weekend, yet very few have thought about securing it. We learn to make things work, not to make them safe.

Consider a college fest registration portal. A student enters their roll number to check their event registrations. The developer, in a hurry to make the feature work, constructs the database query by directly inserting the user's input:

```
query = (
    "SELECT * FROM registrations "
    "WHERE roll_no = '" + roll_no + "'"
)
```

A student entering ' OR '1'='1 instead of their roll number could alter the query's logic, potentially retrieving all registration records. What was intended to be a simple lookup becomes a vulnerability capable of exposing other students' personal information.

The fix is neither complicated nor particularly sophisticated:

```
query = "SELECT * FROM registrations WHERE roll_no = ?"
cursor.execute(query, (roll_no,))
```

The difference is a single programming practice, yet it separates a functional application from a potentially vulnerable one.

The lesson is that security cannot be delegated to the frontend. A registration form may reject malformed inputs through JavaScript, but an attacker can bypass these checks entirely by sending requests directly through tools such as curl or Burp Suite. The backend must independently validate incoming data, use parameterised queries rather than constructing SQL strings, and enforce appropriate access controls.

Building a functional application is only half the responsibility. The other half is ensuring that the application remains safe when someone deliberately tries to break it.

6. A Ladder for Cyber-Safe Education

A ladder that takes every learner below the waterline, at a suitable depth.

Level 1, Awareness (every student and teacher): They must see the real domain before clicking, never share codes or tokens, and know the 1930 helpline and cybercrime.gov.in. Training works: KnowBe4's 2025 benchmark found phishing click rates fell from 33.1% to 4.1% after a year of training.

Level 2, Literate (every graduate): They must understand social engineering, multi-factor authentication and passkeys (A passkey is cryptographically bound to the genuine domain, so a page like discored.com cannot capture one). Passkeys would have stopped the Discord hijack chain completely.

Level 3, Engineer (every technology student): They must follow secure coding, the OWASP Top 10, MITRE ATT&CK and protocol fundamentals, taught alongside web development, not after it.

What we need is for every college to treat cybersecurity awareness more than an occasional advisory and make it a regular part of campus life. Local chapters of organisations such as the

NDLI Club, IEEE Student Branch, IIC and many others can organise hands-on workshops, peer-led sessions, and discussions about the changing landscape of cybersecurity and keeping up with it. They can curate verified learning resources, encourage students to build secure-by-design solutions such as malicious domain detectors, and address the misuse of hijacked accounts for harassment and sextortion. The DPDP Act, 2023, places obligations on institutions concerning the protection of personal data; preparing students to understand and practise cybersecurity is a step towards building a Viksit Bharat that takes those responsibilities seriously.

7. Conclusion

Cyber-safe education is less about teaching students to avoid suspicious links than about building a generation that understands, questions, and takes responsibility for its digital actions. As India advances towards Viksit Bharat, security must become vital to digital education as literacy itself. The goal is not just creating digitally connected citizens, but digitally responsible ones who can build, protect, and sustain a secure digital future.

8. References

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